📖 1. Room Overview

The **How Websites Work** module on TryHackMe is a foundational lab that explains the "magic" behind the browser. It covers how a simple URL request transforms into a fully functional, interactive webpage by exploring the roles of HTML, CSS, JavaScript, and the servers that host them.

🏗️ 2. Task 1: The Three Pillars of the Web

Every website you visit is built using three core technologies that work together:

🦴 HTML (HyperText Markup Language)

- **The Skeleton:** Defines the structure of the page (headings, paragraphs, images).
- **Tags:** Uses elements like `<h1>`, `<p>`, and `<a>` to organize content.

🎨 CSS (Cascading Style Sheets)

- **The Skin:** Controls how the website looks (colors, fonts, layout).
- **Design:** Allows developers to make a site responsive (look good on mobile and desktop).

⚡ JavaScript (JS)

- **The Muscles:** Adds interactivity and logic (buttons, animations, form validation).
- **Client-Side:** Most JS runs directly in your browser, not on the server.

🖥️ 3. Task 2: Client vs. Server

To understand web security, you must distinguish between where the code is running:

- **The Client:** This is **Your Browser**. It receives HTML/CSS/JS and renders it for you to see.
- **The Server:** This is a remote computer that "serves" the files. It often runs **Backend** languages like **PHP, Python, or Node.js** and connects to a **Database** (SQL) to store user data.

🔍 4. Task 3: Viewing the Source

One of the most important skills in web hacking and development is inspecting the code.

- **View Page Source:** Shows the raw HTML sent by the server.
- **Developer Tools (F12):** Allows you to see the "Live" version of the site, including changes made by JavaScript and network requests.

📸 Lab Evidence

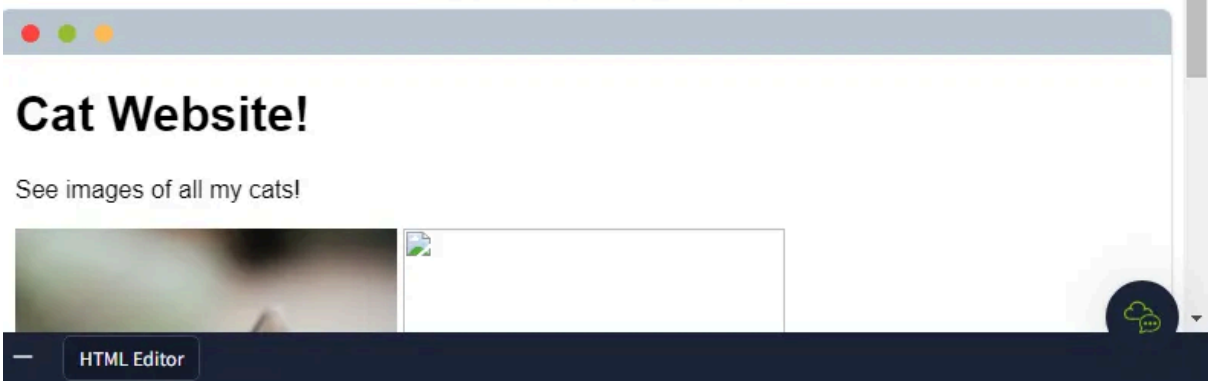
HTML Code

Render HTML Code

```
<!DOCTYPE html>
<html>
    <head>
        <title>TryHackMe HTML Editor</title>
    </head>
    <body>
        <h1>Cat Website!</h1>
        <p>See images of all my cats!</p>
        <img src='img/cat-1.jpg'>
        <img src='img/cat-2.'>
        <!-- Add dog image here -->
    </body>
</html>
```

Type HTML into the box above, then click the "Render HTML" button to see how it looks

Rendered HTML Code



HTML Code

Render HTML Code

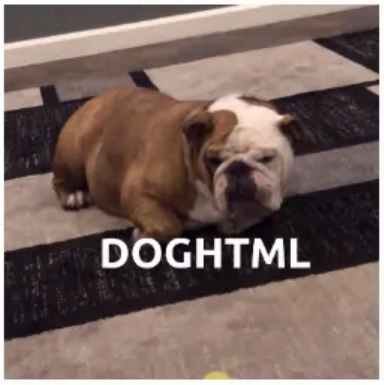
```
<!DOCTYPE html>
<html>
    <head>
        <title>TryHackMe HTML Editor</title>
    </head>
    <body>
        <h1>Cat Website!</h1>
        <p>See images of all my cats!</p>
        <img src='img/cat-1.jpg'>
        <img src='img/cat-2.jpg'>
        <img src='img/dog-1.png'>
    </body>
</html>
```

Type HTML into the box above, then click the "Render HTML" button to see how it looks

Rendered HTML Code

Cat Website!

See images of all my cats!



HTML + Javascript Code

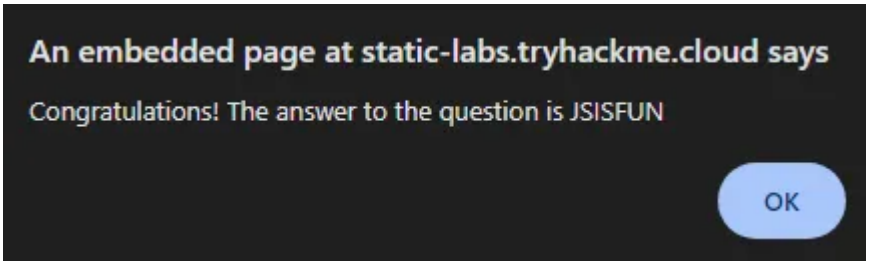
Render HTML+JS Code

```
<!DOCTYPE html>
<html>
    <head>
        <title>TryHackMe Editor</title>
    </head>
    <body>
        <div id="demo">Hi there!</div>
        <script type="text/javascript">
            document.getElementById("demo").innerHTML = "Hack the Planet"
        </script>
    </body>
</html>
```

Type HTML/JS into the box above, then click the "Render HTML+JS" button to see how it looks

Rendered HTML Code

Hack the Planet

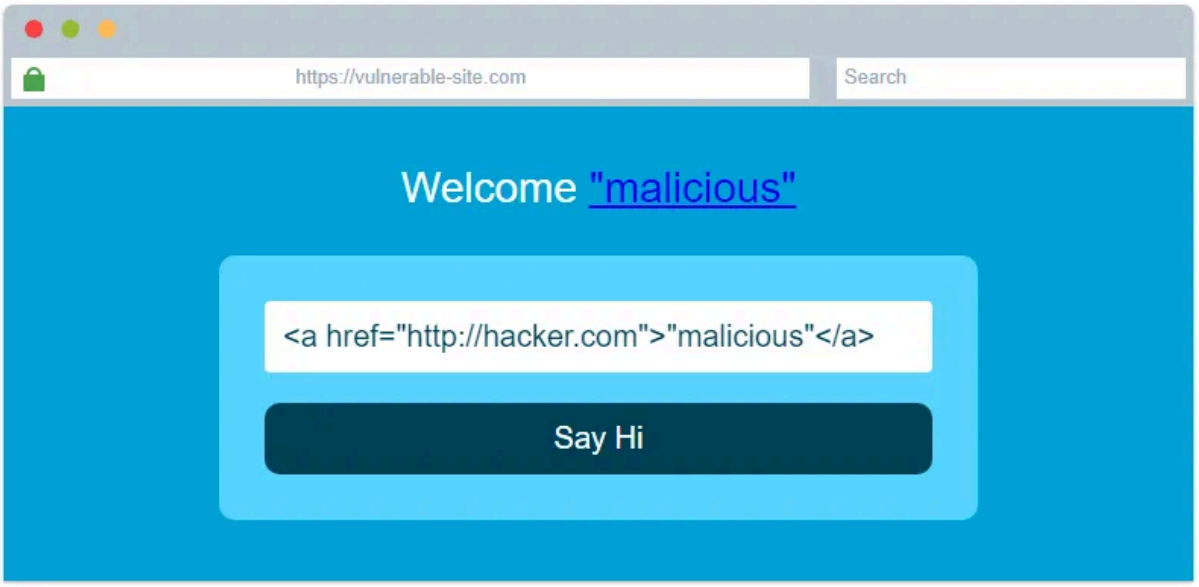


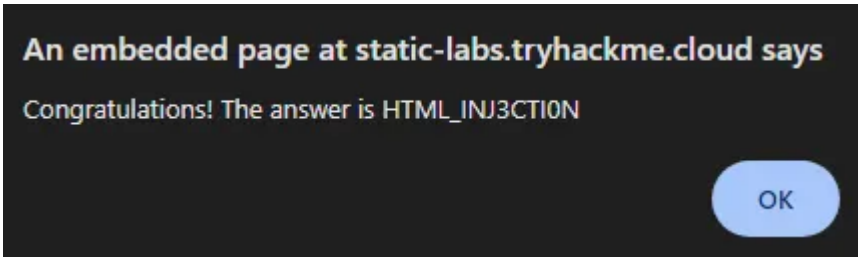
```
Line wrap
<!DOCTYPE html>
<html>

<head>
    <title>How websites work</title>
    <link rel="stylesheet" href="css/style.css"></link>
</head>

<body>
    <div id='html-code-box'>
        <div id='html-bar'>
            <span id='html-url'>https://vulnerable-site.com</span>
        </div>
        <div class='theme' id='html-code'>
            <div class='logo-pos'><img src='img/logo_white.png'></div>
            <p id='login-msg'></p>
            <form method='post' id='form' autocomplete="off">
                <div class='form-field'>
                    <input class="input-text" type="text" name="username" placeholder="Username..">
                </div>
                <div class='form-field'>
                    <input class="input-text" type="password" name="password" placeholder="Password..">
                </div>
                <button onclick="login()" type='button' class='login'>Login</button>
                <!--
                    TODO: Remove test credentials!
                        Username: admin
                        Password: testpasswd
                -->
            </form>
            <div class="footer">Copyright Â© Vulnerable Website</div>
        </div>
    </div>
    <script src='js/script.js'></script>
</body>

</html>
```





🎓 6. Conclusion & Skills Gained

By completing this module, I gained a deep understanding of:

- The relationship between **HTML, CSS, and JavaScript**.
- How to use **Browser Developer Tools** for basic web auditing.
- The difference between **Frontend** presentation and **Backend** logic.